

Influence of Nd_2O_3 on the thermal and dielectric properties of Pb-based lead borosilicate glasses

Byung-Hae Jung, Hyung-Sun Kim *

Department of Materials Science and Metallurgical Engineering, Suncheon National University, 315 Maegok-dong, Suncheon 540-742, South Korea

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Abstract

Neodymium has been employed in various Pb-based glass compositions containing ZnO, MgO, CaO, and BaO, to obtain low melting glasses for use as dielectric materials. Thermal analysis (DTA, TMA) was applied for both qualitative and quantitative thermal properties (T_g , CTE) of the glasses, and impedance analysis of the dielectric properties (ϵ_r , $\tan \delta$) was performed. It was found that with increasing Nd_2O_3 , the glass transition temperature (T_g), the dilatometer softening point (T_d) and dielectric constant (ϵ_r) became higher. Also, a peak shift suggesting neodymium oxide introduced structural changes was found with IR spectra and XPS. These seem to support the conclusion that the addition of increased amounts of Nd_2O_3 can lead to a more rigid glass structure.

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1. Introduction

Recently, both glasses and glass-ceramics containing neodymium oxide, not as a doping element but in significant amounts, have been used for several purposes. Firstly, Nd_2O_3 has been used for special glasses used in halogen lamps that absorb ultra-violet rays which are harmful to humans. Secondly, devitrified glasses containing Nd_2O_3 can have high hardness and excellent chemical durability as refractory glasses [1]. Thirdly, neodymium contained in a thick-film glass layer can be used as a band rejection filter, owing to absorption originating in the intertransition within the 4f shell of the Nd^{3+} ion [2,3]. This last distinct application has been used in several patents for image display devices [4,5].

In a previous study on Nd_2O_3 – B_2O_3 glass systems [11], it was suggested that this glass could be formed in the range from 0 to 28 mol% neodymium oxide, but liquid immiscibility in these systems limits the range for homogeneous glasses to 0–1.5 and 25–28 mol% Nd_2O_3 . In order to overcome this compositional limitation, in other words, to have a stable glass formation region, the substitution of alumina or gallia for boric oxide was

considered to reduce the tendency toward phase separation and stable immiscibility [6]. However, these glasses had high glass transition temperature (T_g) and softening point (T_s) properties. This glass has previously been used for Faraday rotators and other applications, based on its special optical properties [7].

In this study, a different approach was conducted with variation of compositions (PbO – B_2O_3 – SiO_2 – Nd_2O_3 , PbO – B_2O_3 – SiO_2 – ZnO – Nd_2O_3 plus ROs such as MgO, CaO, BaO) in order to obtain low melting glasses that could complete a sintering process at low temperature. The thermal and dielectric influences of Nd_2O_3 were characterized and the structural influence of Nd_2O_3 was carefully considered using X-ray photoelectron spectra (XPS) and infrared ray (IR) spectra. This study should extend not only the thermal, dielectric and structural knowledge of neodymium oxide containing glasses, but also acknowledge the possibility of Nd_2O_3 for use as a potential filter material in plasma display panels.

2. Experimental procedure

All compositions were prepared with chemically pure reagents; PbO, H_3BO_3 , SiO_2 , BaCO_3 , CaCO_3 , MgO, ZnO, and Nd_2O_3 (Table 1). A batch of each

* Corresponding author. Tel.: +82-61 750 3555; fax: +82-61 750 3550.

E-mail address: hskim@suncheon.ac.kr (H.-S. Kim).

Table 1
Compositions

Glass	PbO	B ₂ O ₃	SiO ₂	Al ₂ O ₃	ZnO	CaO	BaO	MgO	Nd ₂ O ₃	Total
G1	65 (35.2) ^a	20 (34.7)	15 (30.2)	–	–	–	–	–	–	100
G2	65 (34.0)	20 (33.5)	15 (29.1)	–	–	–	–	–	10 (3.5)	110
G3	65 (32.8)	20 (32.4)	15 (28.1)	–	–	–	–	–	20 (6.7)	120
G4	65 (32.3)	20 (31.8)	15 (27.7)	–	–	–	–	–	25 (8.2)	125
F1	68 (40.8)	15 (28.9)	5 (11.2)	2 (2.6)	10 (16.5)	–	–	–	–	100
F2	68 (40.0)	15 (28.3)	5 (11.0)	2 (2.6)	10 (16.1)	–	–	–	5 (2.0)	105
F3	68 (39.4)	15 (27.8)	5 (10.7)	2 (2.5)	10 (15.9)	–	–	–	10 (3.8)	110
F4	68 (38.6)	15 (27.2)	5 (10.5)	2 (2.4)	10 (15.6)	–	–	–	15 (5.7)	115
B1	68	15	5	2	–	–	–	10	15	115
B2	60	15	5	2	–	–	–	18	15	115
B3	68	15	5	2	–	10	–	–	15	115
B4	68	15	5	2	–	–	10	–	15	115
B5	60	15	5	2	–	–	18	–	15	115
B6	50	15	5	2	–	–	28	–	15	115

^a wt% with mol% given in parenthesis.

composition, consisting of high purity raw materials was well mixed with a mortar and a pestle. The batches were melted in a platinum crucible at 1100–1200 °C for 1 h, and stirred several times. Each glass melt was quickly poured and quenched into distilled water to obtain glass flakes.

For the preparation of bulk specimens, the glass melts from the furnace were poured into a graphite mold and heated to a temperature of 10 °C above T_g of each glass. The mold then was moved back into the furnace to anneal the glass for 1 h, and then cooled very slowly in the furnace. The glass was removed from the mold and polished to have the required size (0.5×0.5×1 cm) in order to measure the coefficient of thermal expansion (CTE) and dielectric constant. The CTE of the glasses was measured using a vertical type of TMA with a heating rate of 5 °C/min and tested in the temperature range of 25–300 °C. Dielectric properties (ϵ_r , $\tan \delta$) were measured with an impedance phase analyzer. The glass transition temperature (T_g) and crystallization peak were determined with a differential thermal analyzer. Glass powder screened by 500 sieve (<26 µm) was used for IR spectra measurement. The powder was well mixed with KBr powder to make pellets. Bulk glass having 1 mm thickness was used for XPS analysis after polishing with diamond pastes (1 µm).

3. Results

As shown in Fig. 1, the G series (65 PbO·20 B₂O₃·15 SiO₂ plus 0–25 Nd₂O₃ in wt%) gave higher T_g and T_d (dilatometer softening point) values than the F series (68 PbO·15 B₂O₃·5 SiO₂·2 Al₂O₃·10 ZnO plus 0–15 Nd₂O₃ in wt%). Also, T_g and T_d of both series increased with the content of Nd₂O₃. However, the coefficient of thermal expansion was slightly higher in the case of F

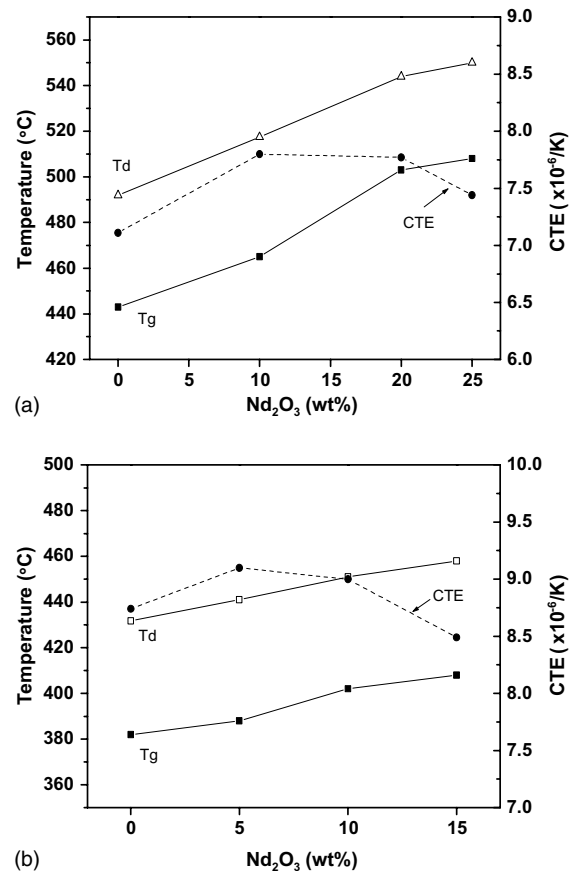


Fig. 1. The glass transition temperature (T_g from DTA), dilatometer softening point and thermal expansion coefficient (T_d , CTE from TMA) as a function of Nd₂O₃ content. (a) G series and (b) F series.

series. The gap of approximately 50 °C difference between T_g and T_d remained constant with increasing Nd₂O₃ in both G and F series, and only a slight change of CTE was shown.

In the B series, other alkali-earth materials were employed in differing amounts. They produced different DTA curves from ZnO containing compositions (F, G series). In Fig. 2(b), compositions containing CaO and BaO had exothermic peaks, indicating crystallization, at relatively low temperatures. Compositions containing MgO showed higher glass transition properties than glass containing ZnO (see F4 and B1 in Fig. 2(a)).

With increasing Nd_2O_3 , relative dielectric constant (ϵ_r) slightly increased from 11.4 to 13.0 at all frequencies. As for the dielectric loss ($\tan \delta$), the value was not affected by composition, as shown in Fig. 3(b). The F series also showed a similar increasing trend of ϵ_r as more Nd_2O_3 was added. However, due to the existence of ZnO in their composition, these compositions had slightly higher ϵ_r than G series; 14.9–16.0 (Fig. 4).

Because of the large number of possible structural groups and complex compositions used in this study, only one network former (B_2O_3) was used for this IR spectra; $68 \text{ PbO} \cdot 20 \text{ B}_2\text{O}_3 \cdot 2 \text{ Al}_2\text{O}_3 \cdot 10 \text{ ZnO}$ plus 0–15 Nd_2O_3 (in wt%). The spectra contain three broad bands in the mid-infrared region at 700 cm^{-1} , between 800 and

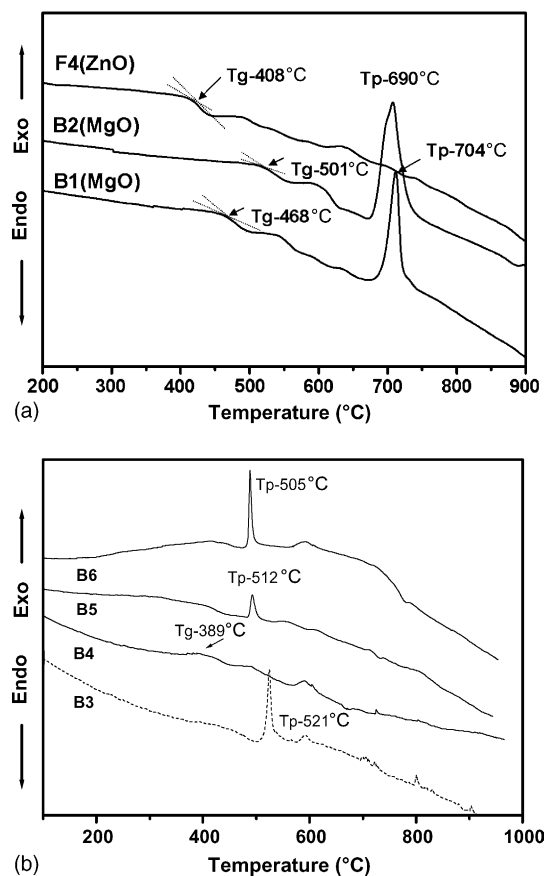


Fig. 2. DTA graphs (T_g , T_p) with different components (B series, $68 \text{ PbO} \cdot 15 \text{ B}_2\text{O}_3 \cdot 5 \text{ SiO}_2 \cdot 2 \text{ Al}_2\text{O}_3 \cdot 10 \text{ MgO}$, CaO, BaO plus 15 Nd_2O_3 in wt%). (a) MgO, ZnO contained and (b) B3 (CaO), B4, B5, B6 (BaO) contained.

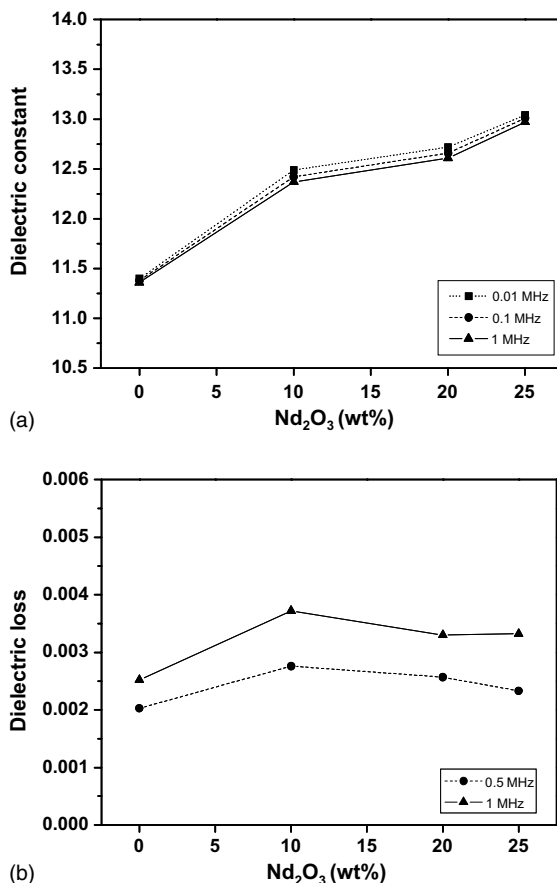


Fig. 3. Dielectric properties with different frequencies (G series): ($\text{PbO} \cdot \text{B}_2\text{O}_3 \cdot \text{SiO}_2 \cdot \text{Nd}_2\text{O}_3$). (a) Dielectric constant (ϵ_r) and (b) loss ($\tan \delta$).

1000 cm^{-1} , and between 1100 and 1500 cm^{-1} (Fig. 5). It is generally accepted that the broad absorption region $1200\text{--}1450 \text{ cm}^{-1}$ is attributed to B–O bond stretching of trigonal BO_3 units, and that in the region of $850\text{--}1100 \text{ cm}^{-1}$ originates from B–O bond stretching of tetrahedral, BO_4 , units [9,10]. The band at 710 cm^{-1} is assigned to bond-bending vibrations of B–O–B linkages of the boron-oxygen network [9–11]. As shown in Fig. 5, the peak at around 700 cm^{-1} became sharper with increasing Nd_2O_3 and shifted down toward higher wave numbers. The specific peaks in both regions (BO_3 , BO_4 units) of these glasses with single B_2O_3 network former were not obvious because of band broadening.

A typical O 1s photoelectron spectrum for $68 \text{ PbO} \cdot 15 \text{ B}_2\text{O}_3 \cdot 5 \text{ SiO}_2 \cdot 2 \text{ Al}_2\text{O}_3 \cdot 10 \text{ ZnO}$ plus 0–10 Nd_2O_3 (in wt%) glasses is found in Fig. 6. The O 1s signal shifts toward higher energy with increasing Nd_2O_3 amount. It was assumed that the O 1s spectrum may be composed of bridging oxygen (BO) and non-bridging oxygen (NBO) components. In Fig. 6, however, there was no noticeable peak distortion, meaning detection for relative concentration of non-bridging oxygen was unsuccessful, even though calculation of full width half maximum (FWHM) was attempted.

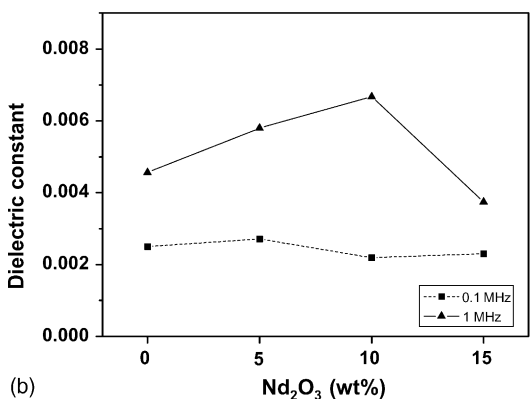
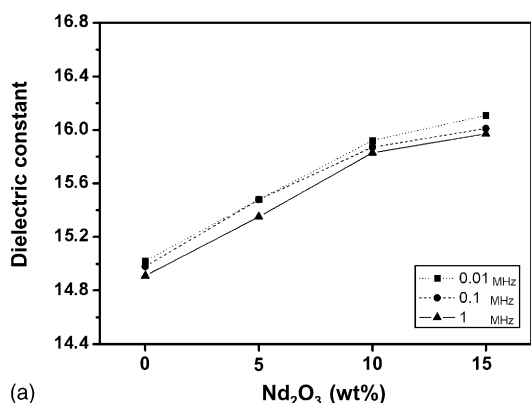


Fig. 4. Dielectric properties with different frequencies (F series). (a) Dielectric constant (ϵ_r) and (b) loss ($\tan \delta$).

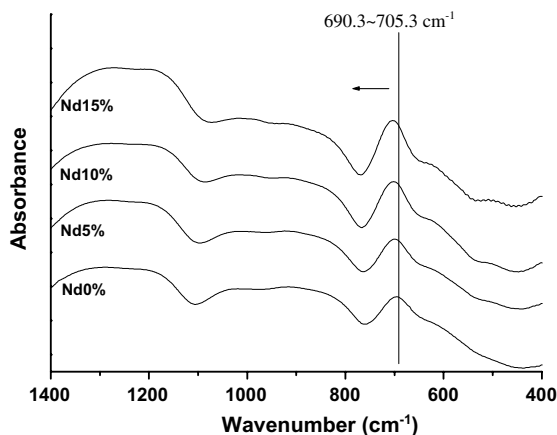


Fig. 5. IR absorbance peak of the glasses which contained only one network former (B_2O_3).

4. Discussion

The merits of glasses containing neodymium oxide have already attracted many glass scientists due to its refractive and optical properties and helium impermeability. Neodymium oxide, a trivalent ion having intermediate characteristics, has been employed in various systems such as aluminosilicate, aluminoborate and al-

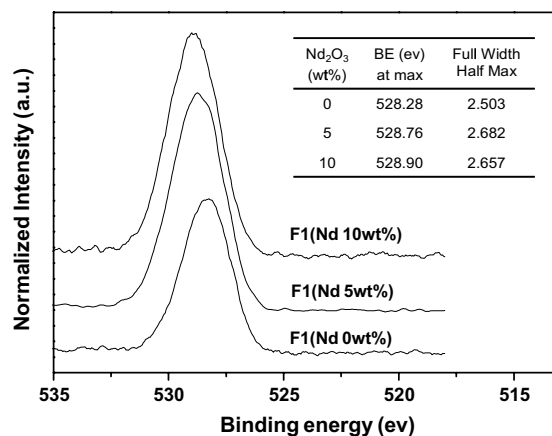


Fig. 6. O 1s binding energy of F series glass with Nd_2O_3 content.

uminogermanate glasses, all of which did not contain any of the oxides defined as glass modifiers and also had high T_g [7,8]. This study employed lead and zinc oxide as the main glass modifiers for a significantly low melting glass system which showed no crystallization at low temperatures. According to the DTA data (Fig. 2), it was found that employing MgO , CaO and BaO could not satisfy these requirements, so the polynary lead-borosilicate and leadborozincsilicate compositions were used for this work.

It is necessary to compare both thermal properties (CTE , T_g , T_d) and dielectric characteristics of the G and F series. As shown in Fig. 1, F series has much lower T_g (about 60 °C) and T_d properties than the G series. The primary reason for this is probably related to the cooperative glass modifying components' role, that is, PbO and ZnO in the case of F series. By adding extra ZnO to replace some SiO_2 as compared with the G series, the F series glasses probably formed more non-bridging oxygen, which is known to decrease T_g . As for the results of the thermal expansion coefficient (experimentally, G series was $7.1\text{--}7.8 \times 10^{-6}/K$, F series was $8.5\text{--}9.1 \times 10^{-6}/K$), a theoretical comparison using the equation discussed by Winkelmann and Schott and Appen was conducted [15]. The equation for calculation of α is given as $\alpha = \alpha_1 P_1 + \alpha_2 P_2 + \dots + \alpha_n P_n = \sum \alpha_i P_i$ where P_i represents the portion of the individual oxides in wt%, and α_i is the characteristic factor for each oxide: PbO 13, B_2O_3 0.33, SiO_2 2.67, Al_2O_3 16.67 and ZnO 6 by Winkelmann and Schott[15]; PbO 13, B_2O_3 -5~0, SiO_2 3.8, Al_2O_3 -3.0 and ZnO 5.0 by Appen [15].

Prior to this study, while thermal factors for other oxides had been determined, the thermal factor of Nd_2O_3 had not yet been determined. Therefore, as an aside, and in order to assist future research in this area, the thermal factor of Nd_2O_3 on the glasses was calculated based on the current results. Unfortunately the factor of Nd_2O_3 is not yet known; therefore the thermal factor of Nd_2O_3 based on the current results. The factor

Table 2
Theoretical thermal factor of Nd₂O₃

Glass	Thermal factor (Nd ₂ O ₃)		CTE ($\times 10^{-6}/\text{K}$)		
	Winkelmann and Schott [15]	Appen [15]	Experiment ^a	Predicted value	
				Winkelmann and Schott	Appen
G2	16.5	24.9	7.8	9.6	10.4
G3	12.8	15.3	7.8	9.6	10.0
G4	10.6	10.3	7.4	9.3	9.2
F2	17.4	23.6	9.1	10.3	10.6
F3	12.9	12.4	9	10.2	10.2
F4	8.2	1.0	8.5	9.7	8.8

^a By TMA from 25 to 300 °C and 5 °C/min of heating rate.

of Nd₂O₃ on the glasses was calculated based on Winkelmann and Schott was found to be approximately 13.1 ± 3.5 , which is similar to the value of PbO, but the value by Appen had somewhat deviating results as 14.6 ± 8.9 (Table 2).

Based on the predicted thermal factor of Nd₂O₃, its value decreased in both series and the predicted thermal expansion coefficient based on the each factor showed very similar decreasing trends, slightly higher than the experimental values. This suggests that the influence of Nd₂O₃ on the thermal factor is more dominant in the glass when the amount is small. A previous study dealing with rare-earth borate glasses [11] was consistent with this theory, showing a decreasing thermal expansion coefficient according to increasing Nd₂O₃, though the current amount of Nd₂O₃ (2–8 mol%) was in a heterogeneous glass region. The different glass formation zone from the previous work was due the difference of compositions, and different ions that were used.

The dominant mechanism contributing to the relative dielectric constant and dielectric losses should be the movement and oscillation of modifying, not forming, ions in their respective interstices under the influence of an alternating electric field [12,13]. Generally the relative dielectric constant and losses of Pb-based compositions with high PbO content similar to the F and G series showed slightly higher values compared to other glass systems such as silicate or borate compositions [13]. A possible reason could be that the lead ion's deformability is much larger than that of barium and zinc ions, even though Pb²⁺ has the second largest ionic size among zinc, barium, neodymium and lead ions ($\text{Ba}^{2+} = 1.42 \text{ \AA}$, $\text{Pb}^{2+} = 1.29 \text{ \AA}$, $\text{Zn}^{2+} = 0.75 \text{ \AA}$, $\text{Nd}^{3+} = 1.04 \text{ \AA}$) [13]. The lead ion's great polarizability ($\text{Pb} = 29.3$, $\text{Ba} = 4.67$, $\text{Zn} = 0.71$, $\text{Nd} = 1.41$) [14] introduces a high deformability. On the other hand, the slight increment in the value of ϵ_r in both series (G and F) as a function of Nd₂O₃ can be ascribed to the higher dielectric constant of the individual Nd₂O₃ in value 19.7 [16] than the Nd₂O₃-less glasses' as 12 (G1) 15 (F1) based on the current results.

In order to disclose the reasons for the thermal and dielectric effects of Nd₂O₃ and to determine whether it is a network former or modifier, or if it has an intermediate role, experimental XPS and IR analysis were used. Increasing T_g and T_d with increasing Nd₂O₃ may have indicated structural changes within the borate groups. The IR spectra of fixed compositions with a single former (B₂O₃) did not show extreme changes in the generally accepted BO₃, BO₄ regions and possible peaks existed inside. However, a noticeable peak at around 700 cm^{-1} , known as a region of B–O–B bending [9–11], shifted toward higher wave numbers. The increased intensity from IR analysis, and the graph shift toward higher binding energy from O1s spectra of XPS (Figs. 5 and 6) probably means that the rigidity of the glasses increased by adding Nd₂O₃. Previous works focusing on lanthanide meta-borate crystals also supports this idea, namely that the $(\text{B}_3\text{O}_6)_\infty$ chains consisting of BO₄ and BO₃ units are interconnected by lanthanum ions, which are situated in distorted 10-cornered polyhedra, to form an infinite chain [9,11]. The possibility of the formation of PbO₄ units which represents the glass forming ability of PbO should also be carefully considered, but the IR spectra (Fig. 5) does not show any vibrational bands between 460 and 475 cm^{-1} which would result from PbO₄ groups [12].

5. Conclusion

The systems PbO–B₂O₃–SiO₂ and PbO–B₂O₃–SiO₂–ZnO were successfully matched with Nd₂O₃ to produce homogeneous glass systems with low glass transition temperatures, without causing crystallization at low temperature. As for thermal influence, Nd₂O₃ raised T_g and T_d , and made the dielectric constant increase slightly. Thermal expansion did not change significantly; in fact, it decreased. Therefore, it was found that Nd₂O₃ as an addition to the PbO–B₂O₃–Al₂O₃–ZnO system, appears to act as a glass former and not simply a modifier.

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